EXPERIMENT 20: TEST FOR ANALGESIA (MOUSE WRITHING ASSAY)

40 STANDARD IN-COURSE MCQs

1. The major purpose of Experiment 20 is to evaluate:

A. Local anesthetic infiltration potency B. Analgesic activity using the mouse writhing assay C. Skeletal muscle relaxant efficacy on isolated phrenic nerve D. Anticonvulsant activity against electroshock

Answer: B

2. What type of noxious pain stimulus is utilized in the mouse writhing assay?

A. Radiant thermal heat B. Chemical visceral pain C. Mechanical tail pressure D. Electrical foot shock

Answer: B

3. The chemical irritant agent used to induce abdominal writhing is:

A. Dilute hydrochloric acid B. Acetic acid (0.6%) C. 5% Formalin D. Capsaicin solution

Answer: B

4. What is the exact concentration of acetic acid specified on Page 112?

A. 0.1% B. 0.6% C. 1.5% D. 5.0%

Answer: B

5. What volume and route are used to administer the acetic acid challenge?

A. 1 mL/kg subcutaneously B. 5 mL/kg orally C. 10 mL/kg intraperitoneally (i.p.) D. 0.1 mL/kg intravenously

Answer: C

6. The experimental animals designated in the manual are:

A. Wistar rats B. Albino mice C. Guinea pigs D. New Zealand rabbits

Answer: B

7. What instrument is used to administer oral solutions to the mice?

A. Oral cannula (gavage needle) B. Tuberculin syringe without needle C. Intravenous butterfly needle D. Glass capillary dropper

Answer: A

8. Which drug serves as the centrally acting opioid analgesic reference standard?

A. Paracetamol B. Morphine sulphate C. Naloxone hydrochloride D. Codeine phosphate

Answer: B

9. What is the dose and route of morphine sulphate in Group III?

A. 1 mg/kg; p.o. B. 5 mg/kg; i.p. C. 10 mg/kg; s.c. D. 20 mg/kg; i.v.

Answer: C

10. Why is morphine administered subcutaneously (s.c.) rather than orally (p.o.) in this protocol?

A. Morphine is insoluble in aqueous solutions B. Morphine has poor and erratic oral bioavailability due to prominent first-pass hepatic metabolism C. Morphine is rapidly destroyed by salivary enzymes D. Subcutaneous administration permanently blocks opioid tolerance

Answer: B

11. Which drug serves as the peripherally acting reference standard?

A. Morphine sulphate B. Acetylsalicylic acid (Aspirin) C. Indomethacin D. Celecoxib

Answer: B

12. What is the dose and route of acetylsalicylic acid in Group II?

A. 10 mg/kg; p.o. B. 50 mg/kg; s.c. C. 100 mg/kg; p.o. D. 200 mg/kg; i.p.

Answer: C

13. What treatment is administered to the control group (Group I)?

A. Normal saline 5 mL/kg; i.p. B. Distilled water 10 mL/kg; p.o. C. 5% Dextrose 10 mL/kg; p.o. D. Liquid paraffin 2 mL/kg; p.o.

Answer: B

14. What is the primary pharmacological role of Group I (Distilled water)?

A. Positive control for opioid analgesia B. Negative vehicle control to establish baseline maximum writhing frequency C. Toxicity control to assess solvent lethality D. Thermal threshold baseline control

Answer: B

15. What are the unknown test substances evaluated in Groups IV, V, and VI?

A. Synthetic local anesthetics B. Plant extracts C. Semisynthetic penicillins D. Inorganic calcium salts

Answer: B

16. What is the dose administered to Group IV (lowest unknown dose)?

A. 10 mg/kg; p.o. B. 25 mg/kg; p.o. C. 50 mg/kg; p.o. D. 100 mg/kg; p.o.

Answer: C

17. What is the dose administered to Group V (intermediate unknown dose)?

A. 50 mg/kg; p.o. B. 100 mg/kg; p.o. C. 150 mg/kg; p.o. D. 200 mg/kg; p.o.

Answer: B

18. What is the dose administered to Group VI (highest unknown dose)?

A. 100 mg/kg; p.o. B. 200 mg/kg; p.o. C. 400 mg/kg; p.o. D. 500 mg/kg; p.o.

Answer: B

19. The doses of the unknown plant extract (50, 100, 200 mg/kg) represent a:

A. Linear arithmetic series (+10 mg/kg) B. 2-fold geometric progression (doubling doses) C. 10-fold logarithmic progression D. Random non-systematic selection

Answer: B

20. How many total treatment groups are included in the experimental protocol?

A. 3 B. 4 C. 6 D. 8

Answer: C

21. What is the waiting time between drug pre-treatment and the acetic acid injection?

A. 5 minutes B. 15 minutes C. 30 minutes D. 60 minutes

Answer: C

22. Why must investigators wait 30 minutes after oral/subcutaneous dosing before injecting acetic acid?

A. To allow animals to recover from handling stress B. To allow sufficient time for drug absorption and achievement of effective systemic concentrations C. To fast the animals completely D. To allow gastric acid secretion to stop

Answer: B

23. A "writhe" is specifically characterized by:

A. Rapid circular running around the cage B. A wave of contraction of abdominal musculature followed by extension of hind limbs C. Clonic seizure convulsions of forelimbs D. Facial grooming and tail biting

Answer: B

24. Over what total duration are writhes counted following acetic acid administration?

A. 10 minutes B. 15 minutes C. 30 minutes D. 60 minutes

Answer: C

25. In what intervals are writhing responses recorded?

A. Every 1 minute for 10 minutes B. Every 5 minutes interval for 30 minutes C. Every 15 minutes for 1 hour D. Continuous single 30-minute block

Answer: B

26. How many 5-minute recording bins are evaluated over the 30-minute test period?

A. 3 bins B. 4 bins C. 6 bins D. 10 bins

Answer: C

27. What is considered evident proof of analgesic efficacy in this assay?

A. An increase in the total number of writhes B. A reduction in the writhing number as compared with the control C. Loss of the righting reflex D. Pupil dilation

Answer: B

28. The mathematical formula for percentage inhibition of writhing is:

A. (Mean drug ÷ Mean control) × 100 B. ([Mean control - Mean drug] ÷ Mean control) × 100 C. ([Mean control - Mean drug] ÷ Mean drug) × 100 D. (Mean control + Mean drug) × 100

Answer: B

29. If control mice produce a mean of 40 writhes and a test group produces a mean of 24.5 writhes, the % inhibition is:

A. 24.5% B. 38.75% C. 50.0% D. 61.25%

Answer: B

30. If control mice produce a mean of 40 writhes and a treated group produces a mean of 35.3 writhes, the % inhibition is:

A. 11.75% B. 25.5% C. 35.3% D. 88.25%

Answer: A

31. If control mice have 40 writhes and aspirin-treated mice show 20 writhes, the % inhibition is:

A. 20% B. 40% C. 50% D. 80%

Answer: C

32. If a drug completely abolishes writhing (0 writhes in all animals), the percentage inhibition is:

A. 0% B. 50% C. 75% D. 100%

Answer: D

33. Intraperitoneal injection of acetic acid triggers pain primarily by causing local release of:

A. Endorphins and enkephalins B. Prostaglandins, bradykinin, and histamine C. Insulin and glucagon D. Acetylcholine and dopamine

Answer: B

34. Which enzyme converts arachidonic acid to pain-sensitizing prostaglandins in the peritoneum?

A. Cyclooxygenase (COX) B. Lipoprotein lipase C. Monoamine oxidase D. Alkaline phosphatase

Answer: A

35. Acetylsalicylic acid (aspirin) reduces writhing by:

A. Stimulating central mu-opioid receptors B. Irreversibly inhibiting cyclooxygenase enzymes to block prostaglandin synthesis C. Blocking voltage-gated calcium channels in the spinal cord D. Inhibiting substance P degradation

Answer: B

36. Morphine reduces writhing by:

A. Inhibiting bacterial wall synthesis B. Activating central opioid receptors in the brain and spinal cord to inhibit pain transmission C. Neutralizing acetic acid chemically in the gut D. Inhibiting cyclooxygenase directly in the peritoneal fluid

Answer: B

37. Why can the mouse writhing assay NOT distinguish between central and peripheral analgesics on its own?

A. Mice do not have a peripheral nervous system B. Both centrally acting opioids and peripherally acting NSAIDs significantly reduce writhing C. Acetic acid only stimulates central brain receptors D. The assay is only sensitive to sedative drugs

Answer: B

38. Which additional test can confirm central analgesic activity over peripheral activity?

A. Hot plate or tail flick test (thermal pain models) B. Righting reflex test C. Blood pressure recording D. Urine glucose assay

Answer: A

39. What type of graph is requested in Question 3 on Page 112?

A. Lineweaver-Burk double reciprocal plot B. % Inhibition vs log dose C. Scatchard binding plot D. Plasma concentration vs time curve

Answer: B

40. Plotting % inhibition against log dose typically yields what type of curve?

A. Bell-shaped Gaussian curve B. Sigmoidal (S-shaped) curve C. Straight vertical line D. Exponential decay curve

Answer: B